INCH-POUND

MIL-T-27/214B
4 May 2012
SUPERSEDING
MIL-T-27/214A
20 April 1993

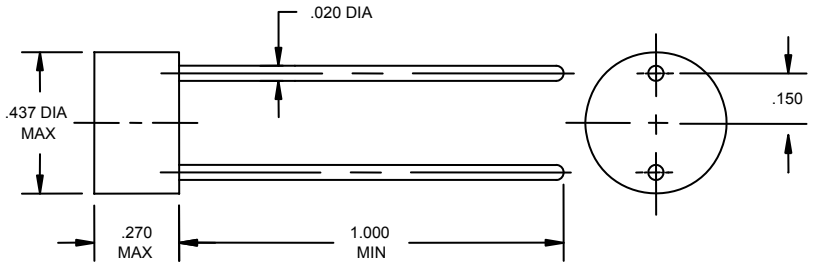
MILITARY SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS (AUDIO, POWER AND HIGH-POWER PULSE), INDUCTORS, AUDIO FREQUENCY

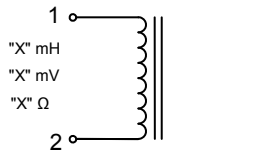
Inactive for new design after 20 April 1993

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.



Inches	mm
.010	0.25
.020	0.51
.150	3.81
.270	6.86
.437	11.10
1.000	25.40



WV-250 V
ALTITUDE-75,000 FT. MAX
(SEE TABLE I FOR VALUE OF "X")

DIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3 Tolerance on terminal position dimensions is ±.010.
4 Marking shall be on the side of the case opposite terminals.
4. Electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations

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REQUIREMENTS: (When numbers in parentheses, such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Working voltage: 250 volts.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Tinned copper, 24 AWG.

Terminal length: 1 inch maximum.

Weight: 2 grams maximum.

Operating temperature range: -55°C to +130°C.

Altitude: 70,000 feet maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 1,000 volts
At reduce barometric pressure: 300 volts

Electrical characteristics (see table I): Power level 0.5 watt maximum.

Temperature rise: 40°C at an ambient temperature of 90°C.

Vibration (high frequency): MIL-STD-202, method 204, test condition D.

VERIFICATION

Qualification inspection: Not applicable.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identification Number (PIN): M27/214-(dash number from table I.)

Supersession data:

MIL-T-27/214 supersedes MIL-T-94(USAF) which is inactive for new design.

Superseding PIN's:

M27/214-08, M27/214-11 and M27/214-14 supersedes M27/94-01, M27/94-02 and M27/94-03, respectively.

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TABLE I. Electrical rating and physical characteristics.

Dash No	Inductance (mH) (1-2)	Voltage (mV) (1-2)	Self-resonant frequency (MHz) (min)	DC resistance (ohms) (1-2) (max)	Temperature coefficient (-55˚C to +130˚C)	Quality factor (min)	Distrib capacitance (pf) (max)
01	0.05 ±2%	11.3	9.000	1.89	±50 PPM	37 at 300 kHz	6
02	0.10 ±2%	16.0	6.000	2.64	±50 PPM	45 at 250 kHz	7
03	0.33 ±2%	29.2	2.800	7.20	±50 PPM	45 at 250 kHz	10
04	0.47 ±2%	34.9	2.300	10.00	±50 PPM	45 at 250 kHz	10
05	0.47 ±2%	13.8	2.200	2.30	±1%	20 at 20 kHz	11
06	1.00 ±2%	50.6	1.500	24.50	±50 PPM	23 at 400 kHz	15
07	1.00 ±2%	20.1	1.100	3.94	±1%	30 at 20 kHz	21
08	1.00 ±2%	13.7	1.800	2.30	±1%	40 at 100 kHz	6
09	3.30 ±1%	92.2	0.630	76.00	±50 PPM	23 at 100 kHz	19
10	3.30 ±1%	36.7	0.720	9.00	±1%	30 at 20 kHz	15
11	3.30 ±1%	25.0	0.860	5.50	±1%	40 at 100 kHz	9
12	10.00 ±1%	160.0	0.310	239.00	±50 PPM	23 at 100 kHz	26
13	10.00 ±1%	43.8	0.390	17.10	±1%	27 at 8 kHz	17
14	10.00 ±1%	65.0	0.475	26.00	±0.25%	60 at 100 kHz	15
15	33.00 ±1%	116.0	0.160	114.00	±1%	28 at 20 kHz	30
16	33.00 ±1%	74.9	0.190	50.20	±1%	14 at 5 kHz	21
17	100.00 ±1%	201.2	0.060	396.00	±1%	14 at 10 kHz	70
18	100.00 ±1%	138.6	0.090	207.00	±1%	14 at 5 kHz	31
19	250.00 ±1%	219.0	0.049	435.00	±1%	14 at 5 kHz	42

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-077)

Review activities:
Army – AR, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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